

Campylobacter transfer from naturally contaminated chicken thighs to cutting boards is inversely related to initial load.



Foods prepared in the kitchen can become cross-contaminated with *Campylobacter* by contacting raw products, particularly skinned poultry. We measured the percent transfer rate from naturally contaminated poultry legs purchased in supermarkets. Transfer of *Campylobacter* from skin ($n = 43$) and from meat ($n = 12$) to high-density polyethylene cutting board surfaces was quantitatively assessed after contact times of 1 and 10 min. The percent transfer rate was defined as the ratio between the number of *Campylobacter* cells counted on the cutting board surface and the initial numbers of *Campylobacter* naturally present on the skin (i.e., the sum of *Campylobacter* cells on the skin and board). Qualitative transfer occurred in 60.5% (95% confidence interval, 45.5 to 75.4) of the naturally contaminated legs studied and reached 80.6% (95% confidence interval, 63.0 to 98.2) in the subpopulation of legs that were in contact with the surface for 10 min. The percent transfer rate varied from $5 \times 10^{-2}\%$ to 35.7% and was observed as being significantly different (Kruskal-Wallis test, $P < 0.025$) and inversely related to the initial counts on poultry skin. This study provides quantitative data describing the evolution of the proportion of *Campylobacter* organisms transferred from naturally contaminated poultry under kitchen conditions. We emphasize the linear relationship between the initial load of *Campylobacter* on the skin and the value of the percent transfer rate. This work confirms the need for modeling transfer as a function of initial load of *Campylobacter* on leg skin, the weight of poultry pieces, and the duration of contact between the skin and surface.